```
%% 1. SYSTEM PARAMETERS (Example 3 - GRN)
A = [0.36 0; 0 0.28];
S = [0.1 0.02; 0.06 0.02];
R = [0.42 0.06; 0 0.3];
Ko = 0; Kq = 1; alpha = 0.01;
exp_term = exp(2*alpha);

% Paper Results (The "Correct" Matrices)
P_sol = [0.9768, -0.0123; -0.0123, 1.0177];
J_sol = [0.8154, 0; 0, 0.8155];
E_sol = [0.5956, 0; 0, 0.6460];

%% 2. MATHEMATICAL VERIFICATION (Proving Phi < 0)
Phi11 = Kq^2*A'*J_sol*A - 2*Kq*Ko*A'*E_sol*A - P_sol;
Phi12 = (Ko+Kq)*A'*E_sol;
Phi13 = Kq*A'*J_sol*R - 2*Kq*Ko*A'*E_sol*R - 0.5*A';
Phi22 = exp_term*P_sol - J_sol - 2*E_sol;
Phi23 = exp_term*P_sol*S + (Ko+Kq)*E_sol*R;
Phi33 = exp_term*S'*P_sol*S + R'*J_sol*R - 2*Kq*Ko*R'*E_sol*R - (R+R')/2;

% Assemble the full 6x6 LMI Matrix
Phi = [Phi11, Phi12, Phi13;
       Phi12', Phi22, Phi23;
       Phi13', Phi23', Phi33];

max_eig = max(real(eig(Phi)));
fprintf('Verification: Max Eigenvalue of Phi is %f\n', max_eig);
```

```
Verification: Max Eigenvalue of Phi is -0.109465
```

```
%% 3. SIMULATION
k_max = 100;
x = zeros(2, k_max);
x(:,1) = [0.4; -0.2]; % Initial concentrations
V = zeros(1, k_max);  % Lyapunov Energy

for k = 1:k_max-1
    % Disturbance w(k)
    if k <= 50
        wk = [0.5*cos(k); 1.8*sin(k)];
    else
        wk = [0; 0];
    end

    % Dynamics
    yk = A*x(:,k) + R*wk;
    fk = max(min(yk, 1), Ko); % Sector [0, 1] nonlinearity
    x(:,k+1) = fk + S*wk;
```

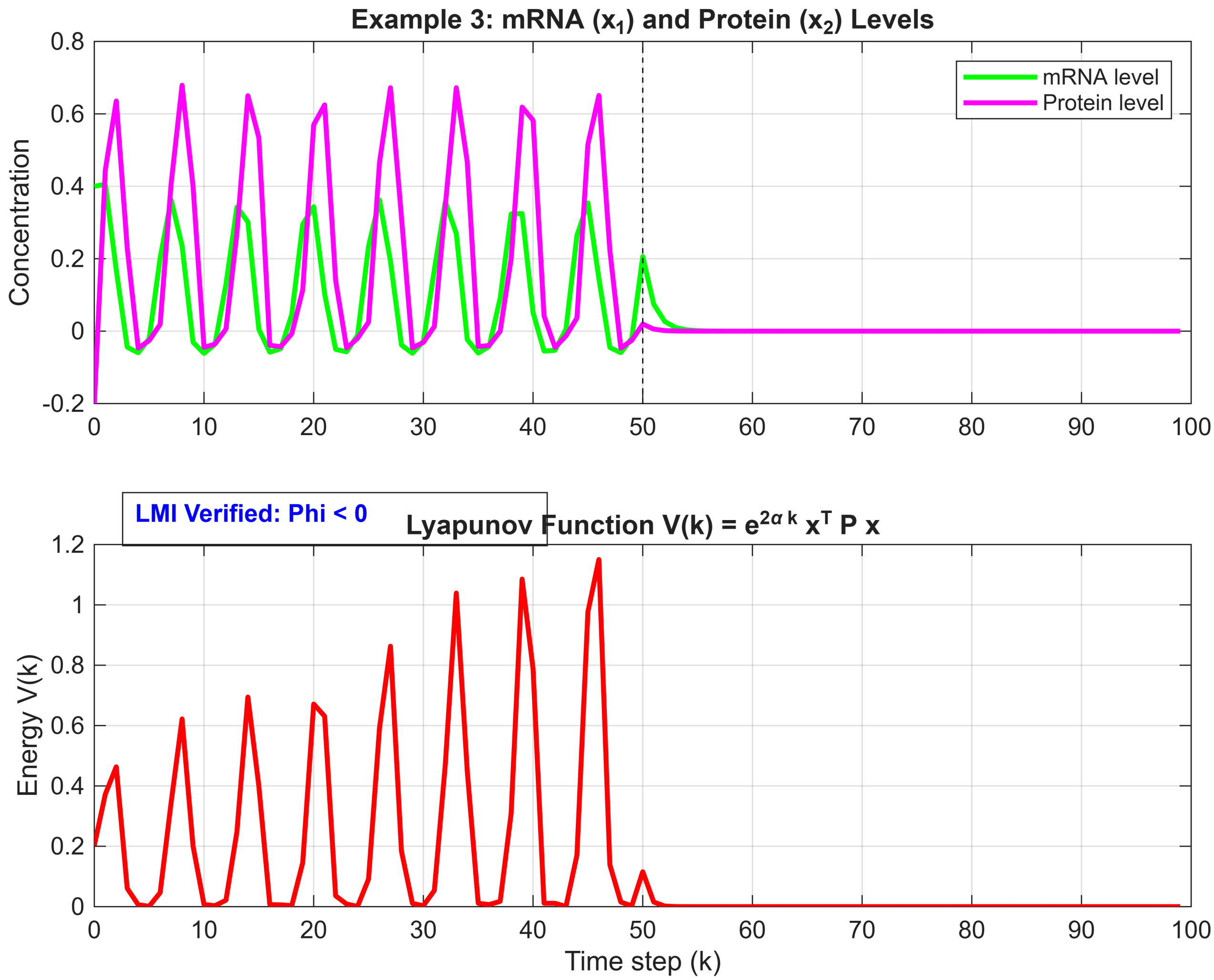
```matlab
    % Calculate Lyapunov Energy V(k) = e^(2*alpha*k) * x'Px
    V(k) = exp(2*alpha*k) * x(:,k)' * P_sol * x(:,k);
end
V(k_max) = exp(2*alpha*k_max) * x(:,k_max)' * P_sol * x(:,k_max);

%% 4. PLOTTING
figure('Color', 'w', 'Position', [100, 100, 800, 600]);

% Subplot 1: State Trajectories
subplot(2,1,1);
plot(0:k_max-1, x(1,:), 'g', 'LineWidth', 2); hold on;
plot(0:k_max-1, x(2,:), 'm', 'LineWidth', 2);
line([50 50], ylim, 'Color', 'k', 'LineStyle', '--');
grid on;
title('Example 3: mRNA (x_1) and Protein (x_2) Levels');
legend('mRNA level', 'Protein level');
ylabel('Concentration');

% Subplot 2: Lyapunov Energy (The Stability Proof)
subplot(2,1,2);
plot(0:k_max-1, V, 'r', 'LineWidth', 2);
grid on;
title('Lyapunov Function V(k) = e^{2\alpha k} x^T P x');
xlabel('Time step (k)');
ylabel('Energy V(k)');

if max_eig < 0
    annotation('textbox', [0.15, 0.45, 0.3, 0.05], 'String', ...
        'LMI Verified: Phi < 0', 'Color', 'blue', 'FontWeight', 'bold');
end
```


Example 3: mRNA (x_1) and Protein (x_2) Levels
Concentration
mRNA level
Protein level
LMI Verified: Phi < 0
Lyapunov Function $V(k) = e^{2\alpha k} x^T P x$
Energy V(k)
Time step (k)